Pandelis Zembashis

CTO | Hands-On Software Engineer | AI Product, Runtime & Infrastructure

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EXECUTIVE SUMMARY

CTO and hands-on software engineer with a record of taking ambiguous, high-leverage technical problems from first principles to shipped systems. At Cosine, lead product and engineering for an applied AI lab building coding agents across runtime, sandboxing, model feedback loops, CLI, desktop, VS Code, cloud, and enterprise deployments. Strong fit for xAI's Exceptional Software Engineer role: broad technical ownership, high initiative, clear communication, and willingness to work on whichever product or infrastructure problem matters most.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – Present

Applied AI lab building autonomous coding agents, model infrastructure, and developer products.

- Lead product and engineering for Cosine's AI coding platform, operating across architecture, implementation, team direction, customer constraints, and product quality.
- Built Cosine's own Go runtime and harness rather than wrapping another agent loop, including tool execution, streaming, checkpointing, deterministic replay, memory, MCP, LSP, and multi-agent orchestration.
- Built sandboxed remote execution from 2024 onward: Kubernetes-scheduled agents, gVisor/runsc isolation, ephemeral per-session environments, network policies, and secure shell/tool access for LLM-controlled coding work.
- Shipped one runtime across CLI, desktop, VS Code/WebAssembly, API-server sessions, and remote Keef pods, keeping product surfaces aligned through one event model, one tool layer, and one release path.
- Reworked filesystem and repository-ingestion hot paths around native Go concurrency, bundled ripgrep-class search, streaming file reads, and low-overhead tool results for large production codebases.
- Built ML feedback infrastructure around the harness: rollout logs, test/LSP/checkpoint signals, human-review outcomes, and parallel sandboxed evaluation runs.
- Supported cloud, private VPC, on-prem, and air-gapped deployments where code, model artifacts, execution logs, and tenant data remain isolated and auditable.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled outbound email and event-driven automation systems that processed millions of email chains and billions of tokens.
- Established engineering cadence, incident response habits, and onboarding rhythms during rapid seed-to-Series A growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a fleet management platform for light electric vehicles in Next.js, React Query, and Tailwind CSS.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving product iteration across web, operations, and marketplace workflows.
- Established development cadences and product intake processes that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021

White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

Substrakt | Software Developer | Aug. 2018 – Jun. 2019
Digital product agency building bespoke publishing, arts, and cultural web platforms.

- Built bespoke WordPress features, search experiences, and internal tooling for design-led client websites.
- Modernized the JavaScript toolchain from Gulp to Webpack, improving development and deployment workflows.

D4 Software | Software Engineer | Jun. 2017 – Jul. 2018
Product engineering consultancy building client platforms, tools, and cloud infrastructure.

- Built client-facing features and internal tools across product and platform engagements using JavaScript and .NET.
- Maintained AWS infrastructure, serverless services, databases, and virtual machines while modernizing the frontend stack with Webpack, TypeScript, and Vue.js.

WRITING & PUBLIC WORK

Why we built our own harness: Technical deep dive on Cosine's Go harness, gVisor-sandboxed Kubernetes substrate, tool execution, replay, multi-agent orchestration, and ML feedback loop.
Cosine Swarm: Long-horizon agent orchestration, parallel work execution, and coordination patterns for AI software engineering.
Tech.eu on Cosine: Coverage of Cosine's benchmark-leading coding models, sovereign AI work, and full-stack AI product strategy.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

AI Runtime & Product Systems: Go, TypeScript, React, WebAssembly, CLI, desktop, VS Code, API servers, agent loops, tool execution
Infrastructure & Sandboxing: Linux, Kubernetes, gVisor/runsc, containers, remote execution, WebSockets, protobuf, AWS, VPC/on-prem/air-gapped deployment
Agent & ML Feedback Loops: Deterministic replay, checkpointing, evaluations, rollout logs, test/LSP signals, reward signals, multi-agent orchestration
Engineering Leadership: High-ownership execution, roadmaps, team cadence, hiring, incident-aware operations, clear technical communication